ثانوية المدينة التعليمية
Education City High School

2.7

Composition of Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Composition order matters: generally $f \circ g \neq g \circ f$.

2. Write one precise sentence explaining this principle: The domain of $f \circ g$ consists of inputs in the domain of g whose outputs lie in the domain of f.

3. Write one precise sentence explaining this principle: Units can reveal meaningful composition order.

4. Write one precise sentence explaining this principle: Decomposition is not always unique.

Core

5. Let $f(x) = \sqrt{x}$ and $g(x) = x - 5$. Find $f \circ g$ and its domain.

6. For $f(x) = 2x + 1$, $g(x) = x^2$, compare $f \circ g$ and $g \circ f$.

7. Decompose $h(x) = \ln(3x^2 + 1)$.

8. State what the formula $(f \circ g)(x) = f(g(x))$ tells you and name any required restriction.

9. State what the formula $\text{Dom}(f \circ g) = \{x \in \text{Dom}(g) : g(x) \in \text{Dom}(f)\}$ tells you and name any required restriction.

10. Create a second representation (table, graph, formula, or context) for one central relationship in Composition of Functions. State its domain and justify one key feature.

Medium

11. Given $f(x) = 2x + 3,\ g(x) = x^2$, compute the indicated natural compositions and domains.

12. Given $f(x) = \sqrt{x},\ g(x) = x - 4$, compute the indicated natural compositions and domains.

13. Given $f(x) = 1/x,\ g(x) = x + 2$, compute the indicated natural compositions and domains.

14. Given $f(x) = \ln x,\ g(x) = x^2 + 1$, compute the indicated natural compositions and domains.

15. Given $f(x) = x^3,\ g(x) = \sqrt[3]{x-1}$, compute the indicated natural compositions and domains.

Hard

16. Find domain of $\sqrt{\frac{1}{x-2}}$. Interpret as a composition.

17. Let f convert Celsius to Fahrenheit and g convert Fahrenheit to Kelvin. Which composition converts Celsius to Kelvin?

18. Decompose $h(x) = \sqrt{5 - 3x^2}$ in two different ways.

19. Find f if $f(g(x)) = 9x^2 + 6x + 2$ and $g(x) = 3x + 1$.

Difficult

20. Find domain of $\ln(\sqrt{x-1} - 2)$.

21. Find all linear f,g with $f \circ g(x) = 6x + 5$, giving one infinite family.

22. Can $f \circ g$ be one-to-one if g is not one-to-one? Prove.

Challenge / HOT

23. Can $f \circ g$ be constant while both f and g are nonconstant? Construct an example.

24. Prove composition is associative.

AP-Style Multiple Choice

25. $(f \circ g)(x)$ means
- **(A)** g(f(x))
- **(B)** f(g(x))
- **(C)** f(x)g(x)
- **(D)** f(x)+g(x)

26. Composition is generally
- **(A)** commutative

(B) not commutative
(C) undefined always
(D) linear

27. Domain of f circle g requires
(A) x in f only
(B) x in g and g(x) in f-domain
(C) all real x
(D) f(x)=g(x)

28. If f(x)=x+1,g(x)=2x, then f circle g is
(A) 2x+1
(B) 2x+2
(C) x+3
(D) 2x^2

29. Outer function in $\sqrt{3x+1}$ can be
(A) $u \mapsto \sqrt{u}$
(B) $x \mapsto 3x+1$
(C) $x \mapsto x^2$
(D) $u \mapsto 1/u$

30. Units help determine
(A) composition order
(B) polynomial degree only
(C) complex zeros
(D) asymptote sign

AP-Style Free Response

31. Let $f(x) = \sqrt{x-1}$ and $g(x) = \frac{1}{x-2}$. (a) Find f circle g and domain. (b) Find g circle f and domain. (c) Explain why they differ.

32. A product is priced in euros by p(q), then converted to QAR by c(e), then taxed by t(r). Write total function, interpret units at each stage, and state domain requirements.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.